Jenna T. B. Ekwealor

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Education

2020 Ph.D., Integrative Biology, University of California, Berkeley. Dissertation: *Tolerance of desiccation and UV radiation in mosses of the genus Syntrichia (Pottiaceae), from genomes to ecology.* Advisor: Brent D. Mishler.

2015 M.S., Environmental Science, California State University, Los Angeles. Advisor: Kirsten M. Fisher.

2012 B.S., Biology, Purdue University, Indianapolis.

2012 B.A., Religious Studies, Indiana University, Indianapolis.

Professional Appointments

2024–present Assistant Professor, Department of Biology, San Francisco State University.

2023–2024 Postdoctoral Fellow in Biology, Department of Biology, Utah State University. Advisor: Carl J. Rothfels.

2021–2023 Biodiversity Genomics Postdoctoral Fellow, Data Science Lab, Smithsonian Institution. Advisor: Rebecca B. Dikow.

2020 Part-time Faculty, Department of Biology, Natural History & Sustainability Program, Merritt College.

Publications

* *contributed equally;* [U] *undergraduate mentee.*

19. Boffa, SC, LR Stark, CM Schine, ML Slate, JC Brinda, JTB Ekwealor, and KK Coe. Recovery from extreme desiccation enhanced by humid air and elevated CO_2 in a widespread desert plant. *Oecologia* (2026). In press.
18. Marks, RA*, RS Abrahams*, and JTB Ekwealor*. Shaped by context: evolutionary trajectories of desiccation tolerance in land plants. *American Journal of Botany* e70180 (2026). DOI: 10.1002/ajb2.70180.
17. Ekwealor, JTB, and KA Simonin. Growth–defence trade-off defines a diffusion limit: a commentary on "Boundary-line trade-off in bryophytes between UV photoprotection and photosynthetic capacity, but not desiccation tolerance." *Annals of Botany* mcag012 (2026). DOI: 10.1093/aob/mcag012.
16. Schneider, AC*, JTB Ekwealor*, A Besik, N Ibrahim, I Ensminger, and S Stefanović. Photosynthetic activity in the heterotrophic plant genus *Cuscuta* (Convolvulaceae) is modulated by phylogeny and ontogeny. *Annals of Botany* mcaf145:1–16 (2026). DOI: 10.1093/aob/mcaf145. Cover feature.
15. Coe, KK, LR Stark, ML Slate, JC Brinda, DN McLetchie, and JTB Ekwealor. Influence of prehydration events on revival of the dryland moss *Syntrichia caninervis* desiccated for 17 years: does water content matter? *The Bryologist* 128(3):487–497 (2025). DOI: 10.1639/0007-2745-128.3.487.
14. Marks, RA, JTB Ekwealor, MAS Artur, L Bondi, TC Boothby, OMS Carmo, DC Centeno, KK Coe, HJW Dace, A Hutt, S Porembski, A Thalhammer, L van der Pas, AJ Wood, P Alpert, D Bartels, S Boeynaems, MN Datar, T Giese, W Ibrahim Seidou, SM Kirchner, J Köhler, UGVSS Kumara, J Kyung, R Lyall, BD Mishler, JBVT Ndongmo Vouffo Epse Tsopze, MS Otegui, V Reddy, J Rexroth, SM Tebele, R VanBuren, J Verdier, UC Vothknecht, MF Wittenberg, E Zokov, MK Oliver, and SY Rhee. Life on the dry side: A roadmap to understanding desiccation tolerance and accelerating translational applications. *Nature Communications* 16:3284 (2025). DOI: 10.1038/s41467-025-58656-y.
13. Ekwealor, JTB, and KM Fisher. Reproduction and population dynamics in autonomous gametophytes. *International Journal of Plant Sciences* 185(4):309–320 (2024). DOI: 10.1086/729606. 2024 Coulter Review; cover feature.
12. Zhang*, X, JTB Ekwealor*, AT Silva, LA Yu, AK Jones, BD Mishler, A Nelson, and MJ Oliver. *Syntrichia ruralis*: Emerging model moss genome reveals a conserved and previously unknown regulator of desiccation in flowering plants. *New Phytologist* (2024). DOI: 10.1111/nph.19620.
11. Dikow, RB, C DiPietro, M Trizna, H BredenbeckCorp, MG Bursell, JTB Ekwealor, R Hodel, N Lopez, WJB Mattingly, J Munro, and R Naples. Developing responsible AI practices at the Smithsonian Institution. *Research Ideas and Outcomes* (2023). DOI: 10.3897/rio.9.e113334.
10. Ekwealor, JTB, and S Roy. Sex chromosomes: How to make a hermaphrodite. *Current Biology* (2023). DOI: 10.1016/j.cub.2023.09.038.
9. Dikow, RB, JTB Ekwealor, WJB Mattingly, MG Trizna, E Harmon, T Dikow, CF Arias, RGJ Hodel, J Spillane, MTN Tsuchiya, L Villanueva, AE White, MG Bursell, T Curry, C Inema, and K Geronimo-Anctil[U]. Let the records show: attribution of scientific credit in natural history collections. *International Journal of Plant Sciences* (2023). DOI: 10.1086/724949.

8. Ekwealor, JTB, SD Benjamin, JZ Jomsky[U], MA Bowker, LR Stark, DN McLetchie, BD Mishler, and KM Fisher. Genotypic confirmation of a biased phenotypic sex ratio in a dryland moss using a novel RFLP technique. *Applications in Plant Sciences* (2022). DOI: 10.1002/aps3.11467. Cover feature.
7. Ekwealor, JTB, and BD Mishler. The transcriptomic effects of acute ultraviolet radiation exposure on two *Syntrichia* mosses. *Frontiers in Plant Science* (2021). DOI: 10.3389/fpls.2021.752913.
6. Ekwealor, JTB, TA Clark, O Dautermann, A Russell, S Ebrahimi, LR Stark, KK Niyogi, and BD Mishler. Natural ultraviolet radiation exposure alters photosynthetic biology and improves recovery from desiccation in a desert moss. *Journal of Experimental Botany* (2021). DOI: 10.1093/jxb/erab051.
5. Silva, AT, B Gao, KM Fisher, BD Mishler, JTB Ekwealor, LR Stark, X Li, D Zhang, MA Bowker, JC Brinda, KK Coe, and MJ Oliver. To dry perchance to live: insights from the genome of the desiccation-tolerant biocrust moss *Syntrichia caninervis*. *The Plant Journal* (2020). DOI: 10.1111/tpj.15116.
4. Ekwealor, JTB, and KM Fisher. Life under quartz: Hypolithic mosses in the Mojave Desert. *PLOS ONE* 15(7):e0235928 (2020). DOI: 10.1371/journal.pone.0235928.
3. Ekwealor, JTB, AC Payton, AE Paasch, KM Fisher, and SF McDaniel. Multiple factors influence population sex ratios in the Mojave Desert moss *Syntrichia caninervis*. *American Journal of Botany* 104(5):1–10 (2017). DOI: 10.3732/ajb.1700045.
2. Meijome, TE*, JTB Ekwealor*, RA Hooker, Y Cheng, WA Ciovacco, SA Balamohan, TL Srinivasan, BR Chitteti, PP Eleniste, MC Horowitz, EF Srour, A Bruzzaniti, RK Fuchs, and MA Kacena. C-Mpl is expressed on osteoblasts and osteoclasts and is important in regulating skeletal homeostasis. *Journal of Cellular Biochemistry* 117:959–969 (2016). DOI: 10.1002/jcb.25380.
1. Eleniste, PP, V Patel, S Posritong, O Zero, H Largura, Y Cheng, ER Himes, M Hamilton, JTB Ekwealor, MA Kacena, and A Bruzzaniti. Pyk2 and megakaryocytes regulate osteoblast differentiation and migration via distinct and overlapping mechanisms. *Journal of Cellular Biochemistry* 9999:1–11 (2015). DOI: 10.1002/jcb.25430.

Preprints

2026 Ekwealor, JTB, SM Kosina, BP Bowen, AT Silva, T Northen, MJ Oliver, and BD Mishler. *Water status shapes responses to ultraviolet radiation in two stress-tolerant mosses*. bioRxiv. DOI: 10.1101/2022.09.14.508022.

2026 Schedler, A, K McCollough, F Rogers, K Macomb, MJ Song, and JTB Ekwealor. *A strong coastal fog gradient produces subtle changes in epiphytic cryptogam communities*. 10.64898/2026.10.16.759761.

2026 Song, MJ, A Schedler, G Wagner, F Freund, IL Maarck, CM Tribble, CJ Rothfels, P Johnson, and JTB Ekwealor. *Molecular data from herbarium specimens reveal a potential extirpation of Azolla microphylla from California*. bioRxiv.

Other Publications

2020 Ekwealor, JTB. A suntan effect in the Mojave Desert moss *Syntrichia caninervis*. *Mojave National Preserve Science Newsletter*, December 2020, 15–19.

Awards & Honors

2026 CSU Chancellor's Early Career Faculty Fellowship.
2020 Biodiversity Genomics Postdoctoral Fellowship, Smithsonian Institution.
2020 Graduate Remote Instruction Innovation Fellowship, University of California, Berkeley.
2019 A.J. Sharp Award Honorable Mention, Botanical Society of America.
2018 Luso-American Education Foundation Scholarship.
2018 Trainee Fellowship Travel Award, International Molecular Moss Science Society.
2017 Pinto-Fialon Fellowship, University of California, Berkeley.
2017 Travel Award, California Botanical Society.
2016 Outstanding Graduate Student Instructor, University of California, Berkeley.
2016 Registration Award, International Workshop on Biological Soil Crusts.
2015 Environment & Society: Data Sciences for the 21st Century, NSF Research Traineeship, University of California, Berkeley.
2015 Berkeley Fellowship, University of California, Berkeley.
2015 Best Presentation Award, Completed Research Category, California Botanical Society.
2015 Best Student Poster Award, California Native Plant Society.
2014 MBRS-RISE MS-to-PhD Graduate Research Fellowship, California State University, Los Angeles.

Research Funding

2026 Gilead Innovation Initiative, Summer Research Award. *Life under snow: Cold tolerance across genomic, physiological, and environmental scales in an alpine moss.* $6,500.

2026–2027 California Species Grant, California Institute for Biodiversity and Experiment Foundation. *Surviving the seasons: Photosynthesis in desert mosses across two desert ecosystems.* $1,365.

2025–2026 California Species Grant, California Institute for Biodiversity and Experiment Foundation. *Tiny worlds in desert moss: Microbes and microfauna in biological soil crusts.* $2,500.

2025 Gilead Innovation Initiative, Summer Research Award. *Genomics of reproduction in a cosexual moss species.* $5,000.

2025–2026 California Species Grant, California Institute for Biodiversity and Experiment Foundation. *Epiphytic cryptogam communities on San Bruno Mountain.* $1,150.

2025–2026 California State University Climate HQ Climate Justice Award. *Environmental degradation, airborne dust, and health inequities in California's Central Valley.* $11,907.

2025–2026 ORSP Affinity Group Catalyst Award. *Environmental degradation, airborne dust, and health inequities in California's Central Valley.* $14,997.

2025–2026 CSUBIOTECH New Investigator Award. *Predicting photosynthetic responses to climate change in California desert mosses.* $14,979.

2020 Myrtle Wolf Grant, California Native Plant Society–East Bay Chapter. $1,400.

2020 Dissertation Award, Department of Integrative Biology, University of California, Berkeley. $2,500.

2020 Paul Silva Student Research Grant, California Botanical Society. $580.

2019 Summer Grant, Department of Integrative Biology, University of California, Berkeley. $3,500.

2019 Research Grant, California Native Plant Society–Bryophyte Chapter. $200.

2019 Grants-in-Aid-of-Research, Sigma Xi Berkeley Chapter. $200.

2018–2019 Mathias Graduate Student Research Grant, University of California Natural Reserve System. $2,000.

2018 Summer Research Award, Department of Integrative Biology, University of California, Berkeley. $1,750.

2017 Anderson & Crum Field Bryology Award, American Bryological and Lichenological Society. $750.

2017 Summer Research Award, Department of Integrative Biology, University of California, Berkeley. $1,750.

2016–2019 Graduate Student Research Allocations Committee Research Award, Department of Integrative Biology, University of California, Berkeley. $1,200 total.

2014 Evo-Devo-Eco Research Exchange Network Grant, NSF Research Coordination Network. $3,000.

Invited Seminars & Conference Presentations

Invited Research Seminars

2026 Deep Springs College, Deep Springs, California.

2026 Department of Biology Retreat, San Francisco State University / California Academy of Sciences.

2025 Center for Comparative Genomics, California Academy of Sciences, San Francisco, California.

2024 Water and Life Interface Institute (WALII) Webinar.

2024 Middlebury College, Department of Biology, Middlebury, Vermont.

2024 Northern California Botanists Symposium, Chico, California.

2024 University of California, Berkeley, University and Jepson Herbaria.

2024 Lewis & Clark College, Natural History Club Moss Appreciation Week Keynote, Portland, Oregon.

2022 San Francisco State University, Department of Biology.

2022 Smithsonian Institution, Smithsonian Botanical Symposium.

2022 University of Tennessee, Knoxville, Department of Ecology and Evolutionary Biology.

2021 University of Colorado, Boulder, Department of Ecology and Evolution.

2021 California Botanical Society, Botany Speaker Series.

2021 University of Hawai'i at Mānoa, Evoluncheon, Ecology, Evolution and Conservation Biology Group.

2021 University of California Botanical Garden, Garden Seminars Program.

Conference Organization and Session Leadership

2026 Session Chair, Biocrust6: Sixth International Workshop on Biological Soil Crusts, Xianyang, China.
2023 *The ploid thickens: methodological developments, empirical advances, and remaining challenges in polyploid phylogenetics.* BOTANY Symposium.
2022 *Stress-tolerant mosses: adaptations to life on the edge, from genes to ecosystems.* BOTANY Colloquium.

Conference Presentations

2026 Biocrust6: The Sixth International Workshop on Biological Soil Crusts, Xianyang, Shaanxi, China. "Mid-Miocene climate change is associated with the evolution of stress tolerance and higher diversification in the biocrust moss *Syntrichia*."
2026 Biocrust6: The Sixth International Workshop on Biological Soil Crusts, Xianyang, Shaanxi, China. "Different defenses under the same sun: responses to ultraviolet radiation in two dryland mosses."
2025 Conference of the American Society of Naturalists, Asilomar, California. "Reproduction and population dynamics in autonomous gametophytes."
2023 BOTANY.
2022 California Native Plant Society Conference (poster); BOTANY (oral).
2021 BL 2021; Southern California Academy of Sciences.
2020 BOTANY.
2019 BOTANY; Biocrust4; International Association of Bryologists.
2018 International Molecular Moss Science Society; American Bryological and Lichenological Society.
2017 California Botanical Society.
2016 BOTANY; Biocrust3.
2015 California Botanical Society; California Native Plant Society.

Teaching Experience

Courses Taught at San Francisco State University

Fall 2026 Plants, People, & Planet (BIOL 529).
Spring 2026 Advanced Biometry (BIOL 710); Advances in Ecology & Systematic Biology (BIOL 862).
Fall 2025 Plant Ecology GWAR (BIOL 529 GW).
Spring 2025 Advanced Biometry (BIOL 710); Advising for Success as a Biology Major (BIOL 231).
Fall 2024 Plant Ecology GWAR (BIOL 529 GW).
Spring 2024 Advanced Biometry (BIOL 710); Skills for Scientific Writing (BIOL 716).

Prior Courses

2020 Instructor, *Natural History of the Bay Area: Bryophytes*, Merritt College.
2020 Graduate Student Instructor, California Natural History, University of California, Berkeley.
2020 Graduate Student Instructor, Integrative Human Biology, University of California, Berkeley.
2020 Graduate Student Instructor, Introduction to California Plant Life, University of California, Berkeley.
2020 Graduate Student Instructor, Thriving in Academia, University of California, Berkeley.
2019 Graduate Student Instructor, Ecosystems of California, University of California, Berkeley.
2015–2016 Graduate Student Instructor, Introductory Biology Laboratory, University of California, Berkeley.
2015 Teaching Assistant, Plant Biology Laboratory for Non-Majors, California State University, Los Angeles.
2014–2015 Teaching Assistant, Ecology Laboratory, California State University, Los Angeles.

Other Teaching and Workshops

2026 Co-Instructor, Science Through Story, Bay Area RaMP Program in Microbiome Sciences.
2026 Co-Instructor, Microbial Genomics, Bay Area RaMP Program in Microbiome Sciences.
2023 Co-Instructor, Using deep learning with digitized herbarium specimen image data, BOTANY 2023.

2023 Certified Instructor, Collaborating and sharing using GitHub without command line, The Carpentries: Data Carpentry, Smithsonian Institution.

2021 Co-Instructor, Wonders of a dryland moss: *Syntrichia* from genomes to ecosystems, University & Jepson Herbaria.

2021 Certified Instructor, The Unix Shell, The Carpentries: Software Carpentry.

2018–2021 Instructor, Biological Sciences Discipline Workshop, Teaching Conference for First-Time GSIs, University of California, Berkeley.

2019 Co-Instructor, Biocrusts: The Living Skin of the Earth, Expanding Your Horizons Girls' Conference.

2018 Helper, Introduction to R, Git, Shell, and Reproducible Analysis in R, The Carpentries.

2018 Helper, Introduction to Shell, Git, and R, The Carpentries: Data Carpentry.

2018 Helper, Genomics Data Wrangling, The Carpentries: Data Carpentry.

Pedagogical and Mentoring Training

2025 Equity-Minded Pedagogy, California State University.

2025 NSF Bay Area Microbiome RaMP Mentoring Training.

2024 Advancing Inclusive Mentoring, California State University.

2020 Certificate of Remote Instruction, University of California, Berkeley.

Fieldwork Experience

2026–present Lead fieldwork across six sites in the Mojave, Sonoran, and Great Basin Deserts of California.

2026–present Co-lead field research on alpine *Syntrichia norvegica* in the Sierra Nevada, California.

2024–2026 Led fieldwork on San Bruno Mountain, California, across coastal fog and inland microclimate gradients.

2023 Co-led fieldwork in the Mojave, Colorado Plateau, and Great Basin deserts of southwestern Utah.

2019–2020 Led fieldwork in the Mojave Desert, California, including installation and retrieval of microclimate loggers (September 2019–February 2020).

2018–2019 Led fieldwork for a year-long Mojave Desert experiment (June 2018–May 2019), including experimental installation and monthly monitoring.

2014 Co-led field sampling of *Syntrichia caninervis* populations for population genetic and reproductive ecology research in the Mojave Desert, California.

Service & Professional Activities

University and Department Service

2026–present Chair, Biology Faculty Communications Committee, SFSU.

2024–present Biology Faculty Greenhouse Ad-Hoc Committee, SFSU.

2024–2026 Biology Faculty Undergraduate Affairs Committee, SFSU.

2024–2026 Biology Faculty Communications Ad-Hoc Committee, SFSU.

2026 University Commencement Faculty Marshal; Biology Graduation Celebration Ceremony.

2024–2026 College of Science and Engineering Student Project Showcase Judge.

2025 Faculty Participant, Explore SFSU.

2022, 2025 Top 100 Outstanding Students Judge, Office of Alumni Relations, Indiana University Indianapolis.

2018–2021 Graduate Student Representative, Department of Integrative Biology Diversity, Equity, & Inclusion Faculty Committee, UC Berkeley.

2018–2020 Graduate Student Representative, Department of Integrative Biology Curriculum Faculty Committee, UC Berkeley.

2018–2019 Graduate Student Representative, Department of Integrative Biology Graduate Student Orientation Committee, UC Berkeley.

2016–2019 Women in Science at Cal Organization Planning Committee, UC Berkeley.

Professional Society Leadership

2024–present President, California Native Plant Society, Bryophyte Chapter.
2018–2024 Student/Post-doc Representative, International Molecular Moss Science Society.
2018–2020 Social Media Chair, California Native Plant Society, Bryophyte Chapter.

Society Memberships

American Bryological and Lichenological Society; American Society of Naturalists; Botanical Society of America; California Botanical Society; California Native Plant Society; International Association of Bryologists; International Molecular Moss Science Society.

Ad-Hoc Reviewer

New Phytologist; *International Journal of Plant Sciences*; *Functional Ecology*; *American Journal of Botany*; *Plant and Soil*; *Annals of Botany*; *The Bryologist*; *Plant Biology*; *Microbial Ecology.*

Public Outreach & Media

2025 NPR Morning Edition / NPR Environment, "Moss Appreciation Week: a lot of celebration for a very little plant."
2024 Atlas Obscura, "Hypolithic Moss."
2021 WTF, Biology? podcast, "Living under a rock has its merits with Dr. Jenna Ekwealor."
2020 Science Friday, "These moss are living their best life–under rocks."
2020 The New York Times, "This moss uses quartz as a parasol."
2020 Berkeley News, "Desert mosses use quartz rocks as sun shades," with subsequent coverage in *The Guardian*, *Smithsonian Magazine*, *Scientific American*, and other outlets.

K–16 Outreach

2026 Insect & Pollination Microscope Activity, Aquatic Park School.
2020 Invited speaker, "10 Coolest Bryophyte Facts," Bronx Arena High School.
2020 Networking guest, "Success Suits You!," Biotech Partners, Berkeley High School.
2019 Invited speaker, Bay Area Science Festival: Celebrating Nature, University & Jepson Herbaria.
2019 Invited speaker, "Introduction to Bryophytes," Plant Systematics, Department of Integrative Biology, UC Berkeley.
2019 Supervisory scientist, Mission Mojave Educational Field Expedition, Blueprint Earth.
2016 Scientist mentor, "Be a Scientist," Community Resources for Science, Willard Middle School.
2016 Logistics volunteer, Expanding Your Horizons Conference, UC Berkeley.